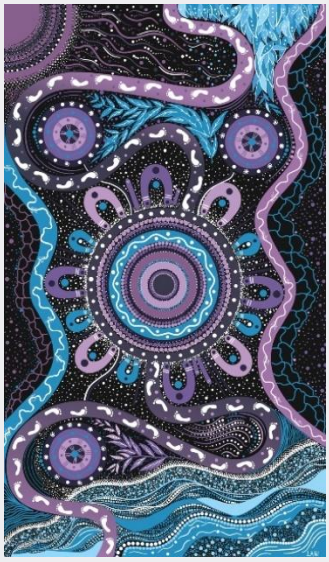
Trip of the Murray 5B2 330 kV busbar on 29 January 2026

May 2026

Reviewable Operating Incident Report under the National Electricity Rules





We acknowledge the Traditional Custodians of the land, seas and waters across Australia. We honour the wisdom of Aboriginal and Torres Strait Islander Elders past and present and embrace future generations.

We acknowledge that, wherever we work, we do so on Aboriginal and Torres Strait Islander lands. We pay respect to the world's oldest continuing culture and First Nations peoples' deep and continuing connection to Country; and hope that our work can benefit both people and Country.

'Journey of unity: AEMO's Reconciliation Path' by Lani Balzan

AEMO is proud to have launched its first Reconciliation Action Plan in May 2024. 'Journey of unity: AEMO's Reconciliation Path' was created by Wiradjuri artist Lani Balzan to visually narrate our ongoing journey towards reconciliation – a collaborative endeavour that honours First Nations cultures, fosters mutual understanding, and paves the way for a brighter, more inclusive future.

Important notice

Purpose

AEMO has prepared this report in accordance with clause 4.8.15(c) of the National Electricity Rules, using information available as at the date of publication, unless otherwise specified.

Disclaimer

To inform its review and the findings expressed in this report, AEMO has been provided with data by registered participants as to the status or response of some facilities before, during and after the reviewable incident, and has also collated information from its own observations, records and systems. Any views expressed in this report are those of AEMO unless otherwise stated, and may be based on information given to AEMO by other persons. AEMO has made reasonable efforts to ensure the quality of the information in this report but cannot guarantee its accuracy or completeness.

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Contact

If you have any questions or comments in relation to this report, please contact AEMO at system.incident@aemo.com.au.

The National Electricity Market (NEM) operates on Australian Eastern Standard Time (AEST). All times in this report are in AEST.

Incident review

This reviewable operating incident[1] report is prepared in accordance with clause 4.8.15(c) of the National Electricity Rules (NER). It has been prepared using information provided by Transgrid[2] and from AEMO systems.

Table 1 Summary of event

	Details
Reviewable operating incident type	Non-credible contingency event impacting critical transmission elements.
Incident details	This report relates to a reviewable operating incident[3] that occurred at 1008 hrs on 29 January 2026 in Victoria. The incident involved the trip of the Murray 5B2 330 kilovolt (kV) busbar.
Incident classification	Other causes – human error. Transgrid protection staff inadvertently connected a test equipment lead to an in-service protection link.
Generation impact	No generation was lost as a result of this incident.
Customer load impact	No load was lost as a result of this incident.
Pre-incident conditions	Immediately prior to the incident, Transgrid protection staff were undertaking protection relay maintenance associated with the in-service Murray 5B1 330 kV busbar. All Murray 1 Power Station (PS) units, all Murray 2 PS units, and the Murray 330 kV No. 1 and No. 3 reactors were out of service prior to the incident.
Incident key events	On 29 January 2026, the following events occurred: 1. At 1008 hrs, the Murray 5B2 330 kV busbar tripped (see Figure 1). 2. At approximately 1025 hrs, AEMO applied hand-dressed ratings[4] to the Dederang – Murray 330 kV No. 67 and No. 68 lines. 3. At 1118 hrs, the Murray 5B2 330 kV busbar was returned to service. 4. At approximately 1120 hrs, AEMO removed the hand-dressed ratings from the Dederang – Murray 330 kV No. 67 and No. 68 lines.
Incident cause	Post-incident investigation by Transgrid confirmed that the Murray 5B2 330 kV busbar tripped due to human error. During protection relay maintenance on the Murray 5B1 330 kV busbar, Transgrid protection staff incorrectly identified a protection link and subsequently connected a test equipment lead to an in-service protection link associated with the Murray 5B2 330 kV busbar. The connection initiated the Murray 5B2 330 kV busbar protection, which operated and tripped the busbar as designed.
Power system response (facilities and services)	There was no other material impact on the broader power system, load or generation.
Rectification	Shortly following the incident, Transgrid restored the Murray 5B1 and 5B2 330 kV busbar protection to normal and ceased all protection work at Murray Switching Station. Following the incident, to mitigate the risk of a similar human error occurring at Murray Switching Station or elsewhere, Transgrid has: 1. Completed a root cause analysis with the protection staff involved. 2. Scheduled a meeting at the end of July 2026 to share the lessons learned from this incident with all Transgrid protection staff undertaking similar protection work.

[1] Reviewable operating incidents are defined by NER 4.8.15(a) and the Australian Energy Market Commission (AEMC) Reliability Panel Guidelines for Identifying Reviewable Operating Incidents.

[2] Transgrid is the owner of Murray Switching Station (MSS).

[3] See NER 4.8.15(a)(1)(i), as the event relates to a non-credible contingency event; and the AEMC Reliability Panel Guidelines for Identifying Reviewable Operating Incidents.

[4] A hand-dressed rating is a rating manually applied by AEMO in its energy management system (EMS) that overrides the static or dynamic ratings.

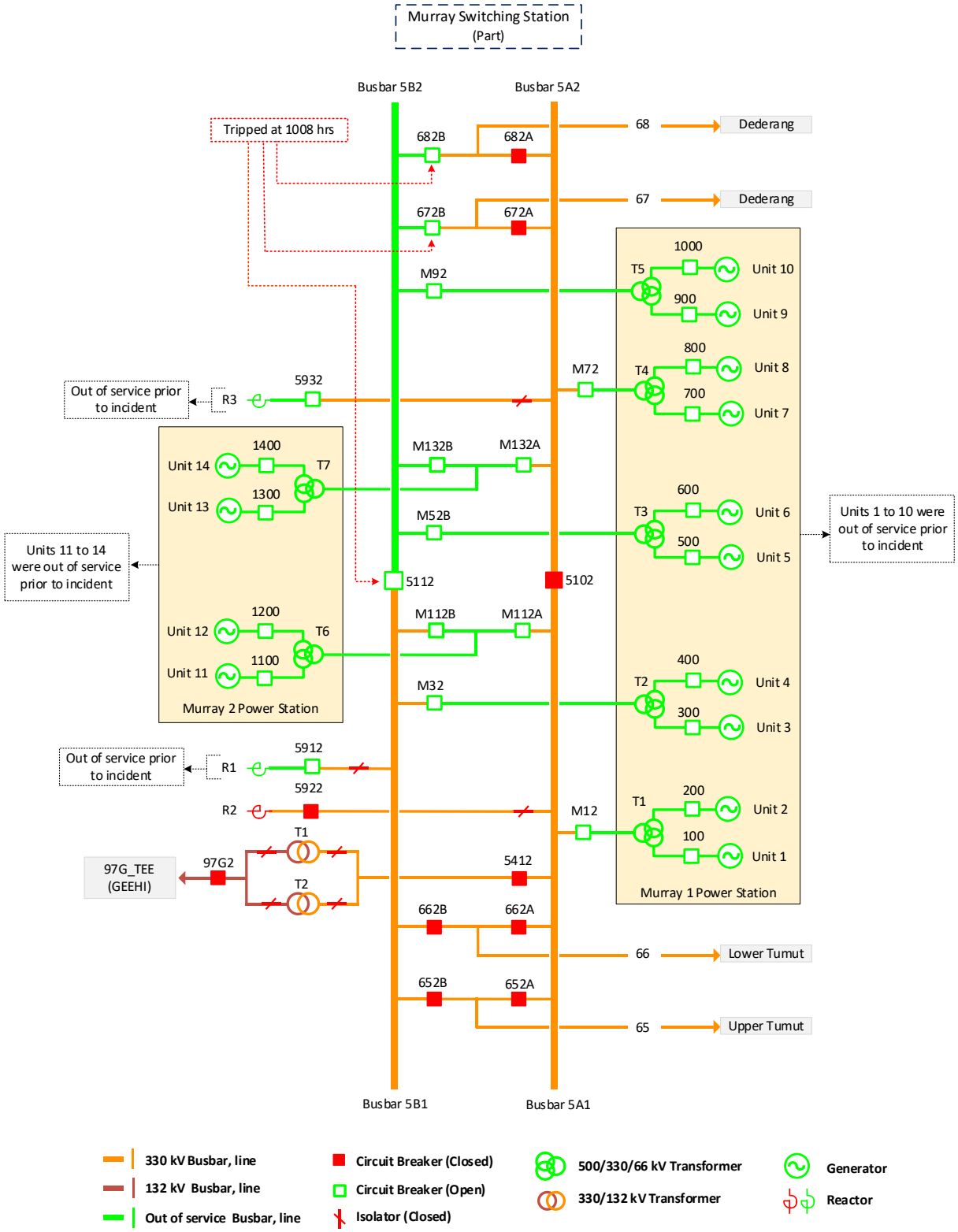
	Details
	3. Implemented a procedural change to use lockout link chocks during protection works at Murray Switching Station to limit the number of accessible elements within protection panels. Transgrid has advised that its other sites already use lockout link chocks, and Murray Switching Station was not due to a different construction making the standard components not fit for purpose at Murray Switching Station.
Power system security	Following the incident, AEMO applied hand-dressed ratings to the Dederang – Murray 330 kV No. 67 and No. 68 lines in accordance with Transgrid's limits advice for single circuit breaker operation at Murray Switching Station. The 15-minute ratings for these lines were reduced from approximately 1370 megavolt amperes (MVA) to 910 MVA. However, this reduced rating did not impact dispatch as network flows remained below the thermal limits and the associated constraints did not bind. Shortly following the restoration of the Murray 5B2 330 kV busbar, AEMO removed the hand-dressed ratings from the Dederang – Murray 330 kV No. 67 and No. 68 lines. The power system remained in a secure operating state throughout this incident and the Frequency Operating Standard (FOS)[5] was met for this incident.
Reclassification	AEMO assessed whether to reclassify this incident as a credible contingency event[6]. Following the incident, on 29 January 2026, Transgrid advised AEMO that the Murray 5B2 330 kV busbar trip was caused by human error. Transgrid further advised that the Murray 5B1 and 5B2 330 kV busbar protection had been restored to normal and protection work at Murray Switching Station had been cancelled. Based on this advice, AEMO was satisfied that another occurrence of this event was not reasonably possible under the current circumstances and did not re-classify it as a credible contingency event. AEMO appropriately applied the reclassification criteria and determined they were not met based on the information available to AEMO at the time.
Market information	For this incident, AEMO issued the following market notices (MN) in accordance with NER requirements: • AEMO issued MN 133160 at 1042 hrs on 29 January 2026 to advise the market of the non-credible contingency event involving the trip of the Murray 5B2 330 kV busbar. • AEMO issued MN 133163 at 1128 hrs on 29 January 2026 to advise the market that the Murray 5B2 330 kV busbar had been returned to service, the cause of this non-credible contingency event had been identified and that AEMO was satisfied the event was unlikely to reoccur under current circumstances.
Conclusions	AEMO has concluded that: 1. At 1008 hrs on 29 January 2026, the Murray 5B2 330 kV busbar tripped inadvertently due to human error. Transgrid protection staff undertaking protection relay maintenance on Murray 5B1 330 kV busbar inadvertently connected a test equipment lead to an in-service protection link associated with the Murray 5B2 330 kV busbar. 2. Following the incident, to reduce the likelihood of a similar human error occurring, Transgrid has implemented a procedural change to use lockout link chocks during protection works at Murray Switching Station to limit the number of accessible elements within protection panels. 3. The power system remained in a secure operating state throughout the incident, and the FOS was met for this incident.
Recommendations	1. AEMO supports Transgrid's plan to share the lessons learned from this incident with all Transgrid protection staff undertaking similar protection work by the end of July 2026. 2. AEMO and Transgrid to share the findings of this incident with the Power System Security Working Group (PSSWG)[7] by Q3 2026.

[5] See https://www.aemc.gov.au/sites/default/files/2023-04/FOS%20-%20CLEAN.pdf.

[6] AEMO is required to assess whether or not to reclassify a non-credible contingency event as a credible contingency event – NER 4.2.3A(c) – and to report how the reclassification criteria were applied – NER 4.8.15(ca).

[7] PSSWG is responsible for providing technical advice to the National Electricity Market Operations Committee (NEMOC) on matters related to the real-time operation of the NEM transmission system.

Figure 1 Post-incident diagram



Abbreviations

Abbreviation	Term
AEMC	Australian Energy Market Commission
AEMO	Australian Energy Market Operator
AEST	Australian Eastern Standard Time
EMS	energy management system
FOS	Frequency Operating Standard
kV	kilovolt/s
MN	market notice
MSS	Murray Switching Station
MVA	megavolt ampere/s
NEM	National Electricity Market
NEMOC	National Electricity Market Operations Committee
NER	National Electricity Rules
PS	Power Station
PSSWG	Power System Security Working Group